



Source-Native Benchmarking for Nature-Inspired Time-Series Families: A Fail-Closed Experimental Protocol

LumenCore Technical Note - Current Public-Safe Review Draft

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This technical note reports local software, custody, benchmark, and protocol states. It is not peer review, independent validation, a performance claim, field validation, trading alpha, realized savings, enterprise value, a patent opinion, or deployment authority.

Abstract

LumenCore registers candidate computational families inspired by natural forms, but does not treat inspiration as evidence. This note reports a source-native benchmark ledger, the retrospective disposition of prior leads, a fixed-rule 12-pair retrospective market panel, and a frozen prospective protocol. The current ledger contains 140 registered families, 35 implementations, and 126 direct candidate-source-baseline comparisons. The market panel repairs the prior single-series bookkeeping bottleneck and contains one narrow exploratory Holm-positive comparison, but its candidate loses on mean to three other registered baselines. No candidate passes the promotion gate. The scientific contribution is therefore a reproducible comparison and falsification framework, not a performance champion.

Research Question

Can a predeclared candidate family beat every accepted baseline for a specific source, series, cadence, and horizon under prospective custody, clustered inference, an effect floor, and familywise error control?

Current Evidence Snapshot

State	Count	State	Count
Registered families	140	Market comparisons inferentially insufficient	48
Implementations present	35	Kraken panel pairs	12
Missing implementations	105	Kraken panel comparisons	16
Direct candidate-source cards	23	Kraken panel exploratory Holm positives	1
Direct comparisons	126	Kraken panel all-baseline mean winners	0
Global Holm-positive comparisons	0	Kraken panel promotions	0
Promoted champions	0	Prospective eligible observations	0
Market-signal comparisons	48		

1. Source-Native Experimental Design

Why source-native baselines matter

A candidate can appear strong when it is compared against an inappropriate cadence, horizon, seasonal period, or baseline. This protocol therefore treats the source contract as part of the hypothesis. A family must win within each named source against the complete accepted baseline roster before any broader promotion.

Evaluation controls

- Candidate-by-source cards evaluated against the baseline roster registered for that source, cadence, series, and horizon.
- Expanding history ends immediately before each forecast origin.
- Inference clusters overlapping origins and horizons at the source-series level instead of treating every row as independent.
- Global Holm correction across the candidate-source-baseline comparison family; prior subset leads receive no preference.
- Positive, neutral, negative, inconclusive, and invalid outcomes are retained.

Registered baseline roster

- naive_last	- drift
- moving_average	- exponential_smoothing
- linear_trend	- seasonal_naive_source_period
- damped_holt_ets	- autoregressive_ridge_source_lag

Retrospective result

The former FRED and TWELVE_DATA subset leads are retired because they do not survive source-series clustering and the complete source-native baseline gauntlet. Four fixed market-signal families were also run against four registered baselines on each of Kraken, TwelveData, and AlphaVantage. All 48 market comparisons are inferentially insufficient because each source currently supplies one source-series cluster. A separate fixed-rule Kraken panel then evaluated the same four candidates against four baselines across 12 pre-scoring-selected pairs. One of 16 comparisons was positive after exploratory global Holm correction: `beast_strategy_trend` versus `ridge_return_baseline`, with mean unannualized risk-adjusted-score delta 0.061277644465, raw exact sign-test p 0.00634765625, and global Holm-adjusted p 0.04443359375. The same candidate lost on mean to buy-and-hold, moving-average-cross, and volatility-targeting baselines. Across the authoritative ledger and panel, zero candidate cards cleared the complete registered baseline set and zero candidates pass promotion.

Current conclusion: no alpha, edge, or champion is established.

2. Frozen Prospective Protocol

Candidate identity

The registry label is **fractal_brownian_surface**. The scientific estimator is **hurst_conditioned_multiscale_increment_heuristic_v1**: Hurst-conditioned weighted multiscale increment heuristic using lags 1, 2, 4, and 8 with a bounded persistence multiplier. The estimator name prevents the registry label from being mistaken for a validated fractional Brownian motion model.

Primary endpoint and multiplicity

The endpoint is `equal_cell_mean_log_relative_mean_absolute_error` ($\theta_{\hat{s},b}$ = arithmetic mean over all fixed cells c in source arm s of $\log(rMAE(c,b))$; every cell has equal weight.). The candidate must clear 16 predeclared contrasts under Holm step-down across all 16 raw one-sided p-values at familywise alpha 0.05. The maximum accepted rMAE is 0.95, corresponding to at least 5 percent relative improvement. Every cell must remain at or below rMAE 1.05, and the arm-level candidate-to-baseline p95 absolute-error ratio must remain at or below 1.10.

Future-only sample gates

- FRED arm: all 12 registered source-series-horizon cells must be complete, with at least 60 joint calendar-month clusters.
- Twelve Data arm: all 3 registered AAPL horizon cells must be complete, with at least 104 joint exchange-week clusters.
- Each arm must retain at least 95 percent of its expected calendar; forward filling, silent row compression, and duplicate timestamps are prohibited.
- Uncertainty: synchronized circular moving-block bootstrap with 20000 frozen replications and a one-sided 95 percent basic-bootstrap upper confidence bound.
- Sample gates cannot be reduced after freeze.

Current protocol state

Protocol	State
LUMENCORE_TS_SOURCE_NATIVE_20260802_V3	SEALED_AWAITING_FUTURE_OBSERVATIONS
Promotion decision	INCONCLUSIVE_WAITING_FOR_NEW_SOURCE_ROWS
Eligible future observations	0

Decision states

- **First-period result**: A bounded first-period result may be reported only after both source arms complete all custody, period, sample, cell, bootstrap, and multiplicity gates. Any incomplete arm leaves the overall result INCONCLUSIVE.
- **Independent replication**: The same unchanged implementation, endpoints, bootstrap, correction family, and decision rule must independently pass in V3_REPLICATION_P2. First-period rows are excluded.
- **Cross-source promotion**: No bounded cross-source promotion claim is allowed before the independent replication period passes.
- **Fail-closed outcomes**: missing external anchor = INVALID_NOT_SCOREABLE; integrity failure = INVALID; insufficient sample or incomplete arm = INCONCLUSIVE; primary failure = CORRESPONDING_BOUNDED_CLAIM_NOT_SUPPORTED.

3. Contribution, Limits, and Reproducibility

Scientific contribution

- A source-native baseline contract that prevents cross-source or cadence-mismatched promotion.
- A custody gate that binds exact source snapshots before benchmark acceptance.
- An append-only prediction-to-settlement chain with immediate external-anchor requests and fail-closed admission until an independent timestamp is verified.
- A clustered inference rule that avoids pseudoreplication from overlapping forecast origins and horizons.
- A full-family multiple-testing rule that prevents cherry-picking isolated wins.
- A future-only protocol with fixed endpoints, effect floor, sample gates, ablations, and falsification states.
- A cost-aware market-signal replay that uses identical timestamps, future-return rows, turnover costs, and source-specific baselines without granting a promotion from descriptive wins.
- A pre-scoring 12-pair Kraken panel that holds candidate, baseline, timing, and 10-basis-point turnover-cost rules fixed while retaining losses and a narrow exploratory positive result.
- A machine-readable claim boundary that keeps software proof separate from field, economic, or deployment claims.

Limitations

- 105 of 140 registered families lack implementations.
- Only 3 lanes currently have executable direct measured adapters; the wider nature-inspired registry remains inventory, synthetic stress, or context until implemented.
- The original 48 market-signal comparisons remain inferentially insufficient under the predeclared five-cluster minimum because each source has one registered series.
- The 12-pair Kraken panel meets the exploratory pair-count floor, but pair-level signs share one exchange and overlapping market timestamps. Independence is therefore unconfirmed, and its one narrow Holm-positive comparison is not confirmatory alpha or edge.
- The panel's narrow trend-versus-ridge result is not a promotion: the same candidate loses on mean to the other three registered baselines, and no candidate clears the complete four-baseline set.
- The prospective protocol has zero eligible future observations and cannot yet support a prospective accuracy conclusion.
- No independent timestamp receipt is present for Version 3; its 15 local seals and pending RFC 3161 query remain non-confirmatory.
- No result establishes universal superiority, field performance, trading alpha, realized savings, customer acceptance, or deployment authority.

Reproducibility receipts

The ledger, frozen protocol, collector, confirmatory analysis, current status, and current analysis receipt are bound below. Hashes establish byte identity and custody only; they do not establish scientific correctness or external acceptance.

Artifact	Bytes	SHA-256
source_native_family_baseline_ledger_latest.json	2244585	627d0add4d31514f7a47dbf8e8c51b0fa46df72635f336133f54abb85b00927a
time_series_source_native_prospective_protocol_v3.json	24737	824a6e6c9d1c22f54705bcab651b1a675d72fce5444864f8659bbd1c04972e8d
time_series_source_native_prospective_collector_v3.py	102303	991f83b8352ffe5f5e5bc4ed4955eb189f9fab468ae36d89fe9a73e0777d3d5d
time_series_source_native_confirmatory_analysis_v3.py	42267	8d52abeff72d990ed05a553644e80c2c54638bf8da224c1b1b181322e0fd116f
TIME_SERIES_SOURCE_NATIVE_PROSPECTIVE_V3_STAT_US_2026-08-04.json	2452	93a0f18d084b9af4f3dd6d53a860514c5cba75708fcff4aeec0b411de56c1c9
confirmatory_analysis_latest.json	1139	56ee284b08877dac59f279cd29ff19377bbbd2a65d89376ca9edd55977b5e519

4. Authorship and Research Integrity

Responsible author

Robert Ashworth, LumenCore. Robert Ashworth is responsible for the research question, protocol ownership, interpretation, release decisions, and all scientific claims.

AI assistance disclosure

Luma (OpenAI Codex) assisted with software implementation, test scaffolding, literature lookup, quality assurance, and document production. AI assistance is not evidence, is not listed as authorship, and does not assume responsibility for the work.

Availability and declarations

Data. Raw provider responses, normalized snapshots, prediction ledgers, and operational receipts are retained locally under append-only custody. Public availability is not claimed.

Code. Canonical code and protocol artifacts are identified by filename, byte count, and SHA-256 receipt. A clean public release remains human-review gated.

Declarations. Funding and competing-interest declarations require responsible-author confirmation before external release.

Method references

- Holm, S. (1979). A Simple Sequentially Rejective Multiple Test Procedure. *Scandinavian Journal of Statistics*, 6(2), 65-70.
- Kunsch, H. R. (1989). The Jackknife and the Bootstrap for General Stationary Observations. *The Annals of Statistics*, 17(3), 1217-1241. <https://doi.org/10.1214/aos/1176347265>
- White, H. (2000). A Reality Check for Data Snooping. *Econometrica*, 68(5), 1097-1126. <https://doi.org/10.1111/1468-0262.00152>
- Hyndman, R. J., and Koehler, A. B. (2006). Another Look at Measures of Forecast Accuracy. *International Journal of Forecasting*, 22(4), 679-688. <https://doi.org/10.1016/j.ijforecast.2006.03.001>

Legacy concept-paper disposition

Three older papers are preserved as historical speculative concepts and blocked from upload. Their BioGeometry, scalar-field, bioresonance, cooling-savings, zero-point, weather-control, and wormhole-adjacent claims are not supported by the current source-native benchmark evidence.

Whitepaper payload SHA-256: 54473978deec35e4fa61c68eb7d5c71521316fab1e1d637b4de11f90fa284aa6